Matemàtiques B

4t ESO

SOLUCIONS DE LA FITXA · UNITAT 3

Equacions i inequacions

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Pots comprovar les equacions i les inequacions amb les eines interactives dels apunts de la unitat.

1 Equacions de segon grau

1. ★☆☆

a) $\boxed{x = \pm 4}$ **b)** $\boxed{x = 0,\ x = -4}$ **c)** $\boxed{x = \pm 5}$

d) No té solució. **e)** $\boxed{x = 0,\ x = \frac{3}{5}}$ **f)** $\boxed{x = \pm\frac{1}{2}}$

2. ★☆☆

a) $\boxed{x = 2,\ x = 3}$ **b)** $\boxed{x = 3,\ x = -5}$ **c)** $\boxed{x = 3,\ x = \frac{1}{2}}$

d) $\boxed{x = 3}$ (doble) **e)** No té solució ($\Delta < 0$). **f)** $\boxed{x = \frac{1}{2},\ x = -\frac{1}{3}}$

3. ★★☆

a) $x^2 + x - 6 = 0$: $\boxed{x = 2,\ x = -3}$ **b)** $x^2 - 5x - 14 = 0$: $\boxed{x = 7,\ x = -2}$

c) $x^2 + 5x - 6 = 0$: $\boxed{x = 1,\ x = -6}$ **d)** $2x^2 - 3x - 9 = 0$: $\boxed{x = 3,\ x = -\frac{3}{2}}$

4. ★★★

a) $x^2 + 9x - 22 = 0$: $\boxed{x = -11,\ x = 2}$

b) $x^2 - 4x + 3 = 0$: $\boxed{x = 1,\ x = 3}$

c) $x^2 - 5x + 4 = 0$: $\boxed{x = 1,\ x = 4}$

d) $x^2 + 3x - 10 = 0$: $\boxed{x = -5,\ x = 2}$

e) $x^2 - x - 6 = 0$: $\boxed{x = -2,\ x = 3}$

f) $x^2 - 2x + 3 = 0$: no té solució

g) $x^2 + 4x + 1 = 0$: $\boxed{x = -2 - \sqrt{3},\ x = -2 + \sqrt{3}}$

h) $x^2 - 2x - 6 = 0$: $\boxed{x = 1 - \sqrt{7},\ x = 1 + \sqrt{7}}$

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5. ★★☆

a) $\Delta = -56$: $\boxed{\text{cap}}$ **b)** $\Delta = 0$: $\boxed{\text{una (doble)}}$ **c)** $\Delta = 5$: $\boxed{\text{dues}}$

$\Delta = 36 - 4k = 0 \Rightarrow \boxed{k = 9}$.

6. ★★★ $(x-3)(x+5) = 0 \Rightarrow \boxed{x^2 + 2x - 15 = 0}$ (o qualsevol múltiple).

2 Equacions biquadrades

7. ★☆☆

a) $t = 9, t = 1$: $\boxed{x = \pm 3, x = \pm 1}$ **b)** $t = 9, t = -4$: $\boxed{x = \pm 3}$

c) $t = -1, t = -2$: no té solució. **d)** $t = 4, t = \frac{1}{4}$: $\boxed{x = \pm 2, x = \pm \frac{1}{2}}$

8. ★★☆

a) $x^2(x^2 - 8) = 0$: $\boxed{x = 0, x = \pm 2\sqrt{2}}$

b) $x^4 + x^2 - 12 = 0$: $\boxed{x = \pm\sqrt{3}}$

c) $\boxed{x = \pm 2}$

9. ★★★

a) $x^4 - 3x^2 - 4 = 0$: $\boxed{x = -2, x = 2}$

b) $x^4 - 5x^2 + 4 = 0$: $\boxed{x = -2, x = -1, x = 1, x = 2}$

c) $x^4 - x^2 - 12 = 0$: $\boxed{x = -2, x = 2}$

d) $x^4 - 10x^2 + 16 = 0$: $\boxed{x = -2\sqrt{2}, x = -\sqrt{2}, x = \sqrt{2}, x = 2\sqrt{2}}$

e) $2x^4 - 9x^2 + 9 = 0$: $\boxed{x = -\sqrt{3}, x = -\frac{\sqrt{6}}{2}, x = \frac{\sqrt{6}}{2}, x = \sqrt{3}}$

f) $4x^2 - 9 = 0$: $\boxed{x = -\frac{3}{2}, x = \frac{3}{2}}$

3 Equacions de grau major que 2

10. ★★☆

a) $\boxed{x = 1, x = 2, x = 3}$ **b)** $\boxed{x = 1, x = -1, x = -2}$

c) $(x-1)^2(x+2) = 0$: $\boxed{x = 1, x = -2}$ **d)** $\boxed{x = 2, x = -\frac{1}{2}, x = -2}$

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11. ★★☆

a) $(x-1)(x+1)(x+2)(x-3)=0$: $\boxed{x=\pm 1,\ x=-2,\ x=3}$

b) $x(x+2)(x-2)=0$: $\boxed{x=0,\ x=\pm 2}$

c) $x(x+1)^2(x-1)=0$: $\boxed{x=0,\ x=-1,\ x=1}$

4 Equacions factoritzades

12. ★☆☆

a) $\boxed{x=4,\ x=-1}$

b) $\boxed{x=0,\ x=-\frac{3}{2},\ x=5}$

c) $\boxed{x=1,\ x=-7}$

d) $\boxed{x=-2,\ x=\frac{1}{4}}$

13. ★★☆

a) $\boxed{x=\pm 4,\ x=-3}$

b) $\boxed{x=1,\ x=2}$ ($x^2+5=0$ no en dona)

c) $\boxed{x=0,\ x=4,\ x=-2}$

d) No és «$=0$»: $x^2-x-6=0$, $\boxed{x=3,\ x=-2}$

5 Equacions amb fraccions algebraiques

14. ★☆☆

a) $\boxed{x=1}$

b) $\boxed{x=3}$

c) $x^2-2x-1=0$: $\boxed{x=1\pm\sqrt{2}}$

d) $x^2-5x+4=0$: $\boxed{x=1,\ x=4}$

15. ★★☆

a) $x^2-3x-10=0 \Rightarrow x=5,\ x=-2$; $x=-2$ anul·la un denominador: $\boxed{x=5}$

b) $x+5=6 \Rightarrow x=1$, que anul·la $x-1$: $\boxed{\text{no té solució}}$

c) $2x^2-3x-2=0$: $\boxed{x=2,\ x=-\frac{1}{2}}$

d) $x^2+4=8 \Rightarrow x=\pm 2$, i totes dues anul·len un denominador: $\boxed{\text{no té solució}}$

e) $x^2-2x-1=0$: $\boxed{x=1\pm\sqrt{2}}$

f) $2(x+1)-3=x$: $\boxed{x=1}$

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16. ★★★ m.c.m. $= 6x(x+1)$: $6(x+1)^2 - 6x^2 = 5x(x+1) \Rightarrow 5x^2 - 7x - 6 = 0 \Rightarrow \boxed{x = 2,\ x = -\frac{3}{5}}$ (totes dues valen).

6 Equacions amb radicals

17. ★☆☆

a) $\boxed{x = 4}$ **b)** $\boxed{x = 13}$ **c)** $\boxed{x = \pm 4}$ **d)** No en té: una arrel no és negativa.

18. ★★☆

a) $x^2 - x - 2 = 0 \Rightarrow x = 2,\ x = -1$; $x = -1$ no val ($\sqrt{1} \neq -1$): $\boxed{x = 2}$

b) $x^2 - 15x + 50 = 0 \Rightarrow x = 5,\ x = 10$; $x = 10$ no val ($10 + 3 \neq 7$): $\boxed{x = 5}$

c) $x^2 - 3x - 4 = 0 \Rightarrow x = 4,\ x = -1$; $x = -1$ no val ($1 + 1 \neq 0$): $\boxed{x = 4}$

d) $x^2 - 10x + 9 = 0 \Rightarrow x = 1,\ x = 9$; $x = 1$ no val ($2 \neq -2$): $\boxed{x = 9}$

e) $x^2 - 2x - 3 = 0 \Rightarrow x = 3,\ x = -1$; $x = -1$ no val: $\boxed{x = 3}$

f) $x^2 - 5x - 6 = 0 \Rightarrow x = 6,\ x = -1$; $x = -1$ no val ($-1 - 3 \neq 2$): $\boxed{x = 6}$

g) $x^2 - x = 0 \Rightarrow x = 0,\ x = 1$; totes dues valen: $\boxed{x = 0,\ x = 1}$

h) $x^2 - 7x + 10 = 0 \Rightarrow x = 5,\ x = 2$; $x = 2$ no val ($3 + 1 \neq 2$): $\boxed{x = 5}$

19. ★★★

a) $x + 7 = 1 + 2\sqrt{x} + x \Rightarrow \sqrt{x} = 3$: $\boxed{x = 9}$

b) $x - 12 = -8\sqrt{x-3} \Rightarrow x^2 - 88x + 336 = 0 \Rightarrow x = 4,\ x = 84$; $x = 84$ no val ($13 + 9 \neq 4$): $\boxed{x = 4}$

7 Equacions exponencials

20. ★☆☆

a) $\boxed{x = 5}$ **b)** $\boxed{x = 5}$ **c)** $\boxed{x = -1}$

d) $\boxed{x = \frac{3}{2}}$ **e)** $\boxed{x = \pm 2}$ **f)** $\boxed{x = -4}$

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21. ★☆☆

a) $\boxed{x = \dfrac{\log 10}{\log 2} \approx 3{,}322}$

b) $\boxed{x = \dfrac{\log 50}{\log 3} \approx 3{,}561}$

c) $\boxed{x = 1 + \dfrac{\log 30}{\log 7} \approx 2{,}748}$

d) $\boxed{x = \dfrac{\log 2}{\log 1{,}05} \approx 14{,}207}$

22. ★★☆

a) $3 \cdot 2^x = 24$: $\boxed{x = 3}$

b) $8 \cdot 3^x = 72$: $\boxed{x = 2}$

c) $\frac{6}{5} \cdot 5^x = 30$: $\boxed{x = 2}$

23. ★★★

a) $t = 2^x$: $t = 1$, $t = 4$; $\boxed{x = 0,\ x = 2}$

b) $t = 3^x$: $t = 1$, $t = 9$; $\boxed{x = 0,\ x = 2}$

c) $t = 4$ ($t = -3$ no val): $\boxed{x = 2}$

8 Equacions logarítmiques

24. ★☆☆

a) $\boxed{x = 1000}$

b) $\boxed{x = 15}$

c) $\boxed{x = 1}$

d) $\boxed{x = \dfrac{26}{3}}$

25. ★★☆

a) $\boxed{x = 5}$

b) $\boxed{x = 3}$

c) $x^2 - 5x + 6 = 0$: $\boxed{x = 2,\ x = 3}$

d) $x^2 + 9x - 10 = 0$; $x = -10$ no val: $\boxed{x = 1}$

e) $x^2 - x - 2 = 10 \Rightarrow x = 4,\ x = -3$; $x = -3$ no val: $\boxed{x = 4}$

f) $x^2 = 9(x - 2) \Rightarrow x^2 - 9x + 18 = 0$; totes dues valen: $\boxed{x = 3,\ x = 6}$

g) $2x + 4 = 3x$: $\boxed{x = 4}$

h) $(x+2)(x-4) = 27 \Rightarrow x^2 - 2x - 35 = 0$; $x = -5$ no val: $\boxed{x = 7}$

26. ★★★

a) $x + 6 = 10(x - 3)$: $\boxed{x = 4}$

b) $x^2 - 2x - 8 = 0$; $x = -2$ no val: $\boxed{x = 4}$

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9 Inequacions de primer grau

27. ★☆☆

a) $\boxed{(4, +\infty)}$



b) $\boxed{[-2, +\infty)}$



c) $\boxed{(-\infty, -4)}$



d) $\boxed{(-\infty, -4]}$



28. ★★☆

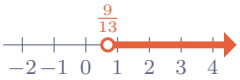
a) $0 \geq 0$, sempre: $\boxed{\mathbb{R}}$



b) $\boxed{(-\infty, 3)}$



c) $\boxed{(\frac{9}{13}, +\infty)}$



d) $-1 > 5$, mai: $\boxed{\varnothing}$

29. ★★☆ $15 + 0{,}5x > 40 \Rightarrow x > 50$: amb $\boxed{\text{més de 50 GB}}$ és més cara la B.

10 Inequacions de segon grau o superior

30. ★★☆

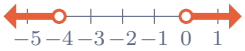
a) $\boxed{(-3, 3)}$



b) $\boxed{(-\infty, -2] \cup [5, +\infty)}$



c) $\boxed{(-\infty, -4) \cup (0, +\infty)}$



d) $\boxed{(-1, 3)}$



e) Només $x = 2$: $\boxed{\{2\}}$



f) $\Delta < 0$ i $a > 0$, sempre positiu: $\boxed{\mathbb{R}}$



31. ★★★

a) $\boxed{(-1, 0) \cup (1, +\infty)}$



b) $\boxed{(-\infty, -2] \cup [1, 4]}$



c) $x(x+1)(x-3) \geq 0$: $\boxed{[-1, 0] \cup [3, +\infty)}$



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32. ★★☆ $x(x+3) < 40 \Rightarrow x^2+3x-40 < 0 \Rightarrow x \in (-8,5)$. Com que $x > 0$: $\boxed{x \in (0,5)}$.

11 Problemes

33. ★★☆ $x(x+1) = 156 \Rightarrow x^2+x-156 = 0 \Rightarrow x = 12$ ($x = -13$ no és natural). Són $\boxed{12 \text{ i } 13}$.

34. ★★☆ $x+y = 17$ i $x^2+y^2 = 169$: $x^2+(17-x)^2 = 169 \Rightarrow x^2-17x+60 = 0 \Rightarrow$ $\boxed{12 \text{ cm i } 5 \text{ cm}}$.

35. ★★☆ $5000 \cdot 1{,}03^t = 6000 \Rightarrow t = \dfrac{\log 1{,}2}{\log 1{,}03} \approx 6{,}17$: calen $\boxed{7 \text{ anys}}$.

36. ★★☆ $\dfrac{600}{x} - \dfrac{600}{x+2} = 25 \Rightarrow x^2+2x-48 = 0 \Rightarrow x = 6$ ($x = -8$ no té sentit): són $\boxed{6 \text{ amics}}$.

37. ★★★ $\dfrac{1}{x} + \dfrac{1}{x+5} = \dfrac{1}{6} \Rightarrow x^2-7x-30 = 0 \Rightarrow x = 10$ ($x = -3$ no té sentit): $\boxed{10 \text{ h i } 15 \text{ h}}$.

38. ★★☆ $x+\sqrt{x} = 42 \Rightarrow \sqrt{x} = 42-x \Rightarrow x^2-85x+1764 = 0 \Rightarrow x = 36,\ x = 49$; $x = 49$ no val ($49+7 \neq 42$): $\boxed{36}$.

39. ★★☆ $x + \dfrac{1}{x} = \dfrac{26}{5} \Rightarrow 5x^2-26x+5 = 0$: $\boxed{x = 5 \text{ o } x = \dfrac{1}{5}}$.

40. ★★★ $x^2 < 3x \Rightarrow x(x-3) < 0$: $\boxed{(0,3)}$



$x^3 > x \Rightarrow x(x-1)(x+1) > 0$: $\boxed{(-1,0) \cup (1,+\infty)}$



41. ★★★ $2^{t/3} = 32 = 2^5 \Rightarrow \frac{t}{3} = 5 \Rightarrow$ $\boxed{t = 15 \text{ hores}}$.

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